BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7465

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Number of chromosomes in human body cells is:

(A) 16
(B) 23
(C) 32
(D) 46

انسان کے جسمانی سیلز میں پائے جانے والے کروموسومز کی تعداد ہے:

Q2. How many hydrogen bonds are present between adenine and thymine?

(A) 1
(B) 2
(C) 3
(D) 4

ایڈینین اور تھائمین کے درمیان کتنے ہائیڈروجن بانڈز ہوتے ہیں؟

Q3. Which is the abiotic component of the ecosystem?

(A) Producers
(B) Herbivores
(C) Air
(D) Carnivores

ایکو سسٹم کا اے بائیوٹک جزو کونسا ہے؟

(A) پروڈیوسرز
(B) ہربی وورز
(C) ہوا
(D) کارنی وورز

Q4. Rabbit is:

(A) Producer
(B) Secondary consumer
(C) Herbivore
(D) Carnivore

خرگوش ہے:

(A) پروڈیوسر
(B) سیکنڈری کنزیومر
(C) ہربی وور
(D) کارنی وور

Q5. When the work on genetic engineering was started?

(A) 1941
(B) 1978
(C) 1990
(D) 1987

جینیٹک انجینئرنگ پر کام کا آغاز کب ہوا؟

Q6. A substance that when absorbed into the body of a living organism and alter normal body function is called:

(A) Antitoxin
(B) Antiseptic
(C) Vaccine
(D) Drug

ایسا مادہ جو جاندار کے جسم میں جذب ہو جانے کے بعد جسم کے نارمل افعال میں تبدیلی پیدا کرے کہلاتا ہے:

(A) اینٹی ٹاکسن
(B) اینٹی سیپٹک
(C) ویکسین
(D) ڈرگ

Q7. Length of trachea is:

(A) 10cm
(B) 12cm
(C) 14cm
(D) 16cm

ٹریکیا کی لمبائی ہے:

Q8. Length of human's kidney is:

(A) 4cm
(B) 5cm
(C) 10cm
(D) 20cm

انسانی گردے کی لمبائی ہے:

Q9. How many layers are of meninges?

(A) 2
(B) 3
(C) 4
(D) 5

مینجیز کی کتنی تہیں ہوتی ہیں؟

Q10. Number of hard bones in an adult's skeleton:

(A) 200
(B) 206
(C) 300
(D) 306

ایک بالغ شخص میں سخت ہڈیوں کی تعداد کتنی ہوتی ہے؟

Q11. Number of chromosomes in endosperm nucleus is:

(A) 1N
(B) 2N
(C) 3N
(D) 4N

اینڈوسپرم نیوکلیس میں کروموسومز کی تعداد ہے:

Q12. Name of individual unit of gynoecium is:

(A) Sepal
(B) Petal
(C) Carpel
(D) Stamen

گائنوشیم کی اکائی کا نام ہے:

(A) سیپل
(B) پیٹل
(C) کارپل
(D) سٹیمن

Biology (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7465

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define bronchioles.

(ii) Write the symptoms of Pneumonia.

(iii) Differentiate between breathing and cellular respiration. (1+1)

(iv) Define osmoregulation and thermoregulation. (1+1)

(v) Write the role of skin to maintain homeostasis.

(vi) How coordination takes place in unicellular organisms?

(vii) Write the functions of midbrain.

(viii) Write definition of acromegaly.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) What are the types of connective tissues?

(ii) Define flexion and extension.

(iii) What is placenta?

(iv) Define dormancy and give its importance.

(v) Write the names of nitrogenous bases of DNA.

(vi) Differentiate between genotype and phenotype.

(vii) What is codominance?

(viii) State the Mendel's Law of Segregation.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define population.

(ii) What is the primary source of energy in an ecosystem?

(iii) Define biomass.

(iv) Differentiate abiotic and biotic components.

(v) Define genetic engineering.

(vi) What is a fermenter?

(vii) Define addictive drug.

(viii) Which antibiotic was discovered by Fleming?

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5. (Marks 09) Describe the function of kidney in detail.

Q6. (Marks 09) Define arthritis. Also describe its types.

Q7. (Marks 09) Define neuron and write a complete note on neuron.

Q8. (Marks 09) Define fertilization and explain its types.

Q9. (Marks 09) Explain the basic steps in genetic engineering.

Q10. (Marks 09) What are antibiotics? Describe three major groups of antibiotics.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7468

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Growing of an entire new plant from part of original plant is called:

(A) Budding
(B) Regeneration
(C) Fragmentation
(D) Vegetative propagation

پودے کے کسی حصے سے مکمل نیا پودا بنانا کہلاتا ہے:

(A) بڈنگ
(B) ری جنریشن
(C) فریگمنٹیشن
(D) ویجیٹیٹو پروپیگیشن

Q2. Transfer of pollen grain from anther to stigma is called:

(A) Reproduction
(B) Pollination
(C) Fertilization
(D) Respiration

اینتھر سے پولن گرین کی سٹگما تک منتقلی کہلاتی ہے:

(A) ری پروڈکشن
(B) پولینیشن
(C) فرٹیلائزیشن
(D) ریسپائریشن

Q3. Human arm has bones:

(A) 8
(B) 14
(C) 3
(D) 22

انسانی بازو میں ہڈیاں ہیں:

Q4. Which is not a layer of eye?

(A) Pupil
(B) Sclera
(C) Choroid
(D) Retina

ان میں سے کونسی آنکھ کی تہہ نہیں ہے؟

(A) پیوپل
(B) سکلیرا
(C) کورائیڈ
(D) ریٹینا

Q5. Normal human temperature is:

(A) 37°C
(B) 98°C
(C) 100°C
(D) 0°C

انسان کا نارمل درجہ حرارت ہے:

Q6. The length of trachea is:

(A) 10cm
(B) 12cm
(C) 14cm
(D) 16cm

ٹریکیا کی لمبائی ہے:

Q7. Which drug is obtained from plants?

(A) Aspirin
(B) Opium
(C) Cephalosporin
(D) Insulin

کونسی دوا پودوں سے حاصل ہوتی ہے؟

(A) ایسپرین
(B) افیون
(C) سیفالوسپورن
(D) انسولین

Q8. In which process there is incomplete oxidation-reduction of glucose?

(A) Genetic engineering
(B) Photosynthesis
(C) Fermentation
(D) Biotechnology

کس عمل کے دوران گلوکوز کی نامکمل آکسیڈیشن ریڈکشن ہوتی ہے؟

(A) جینیٹک انجینئرنگ
(B) فوٹو سنتھیسز
(C) فرمنٹیشن
(D) بائیوٹیکنالوجی

Q9. Enrichment of water with inorganic nutrients is called:

(A) Eutrophication
(B) Contamination
(C) Fertilization
(D) Respiration

پانی میں ان آرگینک غذائی مادوں کا اضافہ کہلاتا ہے:

(A) یوٹروفیکیشن
(B) آلودگی
(C) فرٹیلائزیشن
(D) ریسپائریشن

Q10. Base of food chain is:

(A) Producer
(B) Consumer
(C) Decomposer
(D) Virus

فوڈ چین کی بنیاد ہے:

(A) پروڈیوسر
(B) کنزیومر
(C) ڈی کمپوزر
(D) وائرس

Q11. How many contrasting traits Mendel studied in pea plant?

(A) 17
(B) 7
(C) 3
(D) 4

مینڈل نے مٹر کے پودے کی کتنی متضاد خصوصیات کا مطالعہ کیا؟

Q12. DNA wraps around histone protein and form:

(A) Chromosome
(B) Spindle
(C) Nucleosome
(D) Allele

DNA ہسٹون پروٹین کے گرد لپٹ کر بناتا ہے:

(A) کروموسوم
(B) سپنڈل
(C) نیوکلیوسوم
(D) ایلیل

Biology (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7468

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Write the names of two phases of breathing.

(ii) Write the symptoms of bronchitis.

(iii) What is nicotine?

(iv) What is the difference between thermoregulation and osmoregulation?

(v) Define lithotripsy.

(vi) What is meant by nerve impulse?

(vii) Write the functions of insulin and glucagon.

(viii) Owl is unable to see during day time. Give reason.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) What are chondrocytes?

(ii) Define skeleton.

(iii) What is grafting?

(iv) Define pollination.

(v) Define genotype.

(vi) Define Mendel's Law of Segregation.

(vii) What is artificial selection?

(viii) What is the effect of dominant allele on recessive allele?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define ecosystem and biosphere.

(ii) What is meant by acid rain?

(iii) What is eutrophication?

(iv) How the flow of energy is different from that of materials?

(v) Define genetic engineering.

(vi) Define fermentation with reference to biotechnology.

(vii) What is the difference in pharmaceutical drug and addictive drug?

(viii) Write any two uses of tetracyclines.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5. (Marks 2+2+1) What are hydrophytes? Describe osmotic adjustments in xerophytes and halophytes.

Q6. (Marks 3+1) What is arthritis? Describe its types.

Q7. (Marks 05) Describe the structure of the external, middle and inner ear.

Q8. (Marks 04) Describe the necessary conditions for the germination of seeds.

Q9. (Marks 2+2+1) What is biotechnology? Describe its applications in the fields of medicine and environment.

Q10. (Marks 4×1) Describe any four sources of medicinal drugs.